

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

- AI Engineer - Fedway Associates**, EliteUS Software Solutions Aug 2024 - Present | Edison, NJ
- Designed and deployed GenAI text/vision systems using LLMs and RAG with semantic embeddings and LangChain pipelines, improving response accuracy by 30%+ and reducing retrieval latency by 25%.
 - Built production-grade GenAI chatbots using AutoGen, RAG pipelines, and vector databases, enabling multi-agent reasoning and human-in-the-loop escalation while improving response accuracy by 30%+.
 - Built recommender systems from raw sales data, analyzing customer purchase behavior, creating customer profiles and interactive dashboards that increased customer engagement by 30% and supported data-driven sales analysis.
- Data Science Intern**, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida
- Led an ads analytics initiative to build scalable data pipelines and monitoring systems, enabling visibility from 15% to 45% in impression delivery, pacing behavior, and campaign performance across large-scale datasets.
 - Engineered ML workflows using H2O AutoML, XGBoost, and FastAPI NLP services, improving sentiment analysis and engagement patterns accuracy by 65% and campaign spend efficiency by 50%.
 - Built analytics and evaluation systems combining SQL pipelines, forecasting models and automated validation, improving engagement quality by 40% and supporting data driven audience targeting.
- Application Development Associate**, Accenture June 2021 - Aug 2022 | Hyderabad, India
- Optimized ETL workflows and Python data pipelines with CI/CD integration, reducing support calls by 50%.
 - Performed analytics on millions of records using SQL, AWS Athena, Redshift, leveraging pandas for data cleaning and feature preparation while visualizing insights through interactive dashboards in Tableau and Power BI.
 - Increased system reliability by 25% through optimization, performance tuning using SQL Workbench, while collaborating with cross-functional teams translating business needs into reporting solutions.

TECHNICAL SKILLS

Programming: Python, R, Scala, SQL. **Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, FastAPI

Machine Learning: Supervised & Unsupervised Learning, Classification, Clustering, Ensembling, Feature Engineering.

GenAI & LLMs: LLM fine-tuning, RAG, Prompt Engineering, LangChain, Vector Databases, Semantic Search.

Cloud & MLOps: PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Pinecone, Chroma.

Deep Learning: Neural Networks, CNNs, LSTMs, YOLO, Faster R-CNN, GANs, Time-Series Forecasting.

Statistical Analysis: Linear Regression, Bayesian Inference, Hypothesis Testing, Causal Inference, Model Evaluation.

EDUCATION

- Rutgers University**, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ
- Relevant Coursework:** Statistics, Neural Networks, Machine Learning, Data Structures and Algorithms(Python), Data Wrangling (R), NLP, Data Mining. Personal Website: <https://jayanthdasamantharao.github.io/>
- Andhra University** B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

- Self Driving Cars using Image Processing and Deep Learning** Developed a self-driving car pipeline using CNNs, YOLO, and Faster R-CNN with sensor fusion (camera, LiDAR, radar), achieving 95%+ object detection accuracy and enabling real-time inference in a Unity3D + Python/Flask simulation.
- Bayesian Classification** - [Bayesian Networks, PyMC3, Informative Prior Distributions] Developed Bayesian logistic regression models using PyMC3 with informative priors to predict diabetes status, achieving 88% recall and 0.85 F1-score, and comparing performance across uniform and normal prior distributions.
- Language Identification** - [Naive Bayes, Logistic Regression, BERT, embeddings] Worked on identifying languages using Naive Bayes, Logistic Regression, and DistilBERT, incorporating n-grams analysis achieving 96% accuracy.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>